Method 2:

The HCF of 48 and 84 is 12 (Find it!). Thus, we can get its standard form by dividing it by –12.

$$\frac{48}{-84}=\frac{48\div(-12)}{-84\div(-12)}=\frac{-4}{7}$$

(ii) **Method 1:**

$$\frac{-18}{-42}=\frac{-18\div(-2)}{-42\div(-2)}=\frac{9\div3}{21\div3}=\frac{3}{7}$$ (dividing by –2 and 3 successively)

Method 2:

The HCF of 18 and 42 is 6 (Find it!). Thus, we can get its standard form by dividing it by 6.

$$\frac{-18}{-42}=\frac{-18\times(-1)}{-42\times(-1)}=\frac{18}{42}=\frac{18\div6}{42\div6}=\frac{3}{7}$$



Try these

1. Which of the following pairs represents equivalent rational numbers?

   (i) $\frac{-6}{4}, \frac{18}{-12}$ (ii) $\frac{-4}{-20}, \frac{1}{-5}$ (iii) $\frac{-12}{-17}, \frac{60}{85}$

2. Find the standard form of:

   (i) $\frac{36}{-96}$ (ii) $\frac{-56}{-72}$ (iii) $\frac{27}{18}$

3. Mark the following rational numbers on a number line.

   (i) $\frac{-2}{3}$ (ii) $\frac{-8}{-5}$ (iii) $\frac{5}{-4}$

1.2.6 Comparison of rational numbers

It is useful to remember the following points:

- Every positive number is greater than zero.
- Every negative number is smaller than zero.
- Every positive number is greater than every negative number.
- Every number on the right of a number on a number line is greater than that number.

When two integers or fractions are given, we know how to compare them and say which is greater or smaller. Now, in the same way, we can compare a pair of rational numbers.

Type 1 : Comparing two rational numbers with opposite signs

Example 1.3

Compare $\frac{5}{17}$ and $\frac{-10}{19}$.

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